

Network structure

Initial conditions

1. L - The amount of training data
2. M - The size of the input layer
3. N - The size of the inner layer

Input

Position - a set of combinations (cell, type, color)

Cell

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Cell \in \{0, \dots, 63\}
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Type

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Type \in \{0, \dots, 5\} \\\
Pawn = 0, \ Knight = 1, \ Bishop = 2, \ Rook = 3, \ Queen = 4, \ King = 5
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Color

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Color \in \{0, 1\} \\\
White = 0, \ Black = 1
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The Chinese Remainder Theorem

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(\mathbb{N}/_{64\mathbb{N}}; \ \mathbb{N}/_{6\mathbb{N}}; \
\mathbb{N}/_{2\mathbb{N}}) \simeq \mathbb{N}/_{768\mathbb{N}}
```

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(color, \ type, \ cell) \hookrightarrow (color \times 64 \times 6) + (type
\times 64) + cell
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